

# Quality Inspection Report QIR-2025-0233

Dimensional / Lot Acceptance Inspection | SLEEVE, INNER, SHIFTING (QPQ nitrided)

|                        |                                                                              |
|------------------------|------------------------------------------------------------------------------|
| Report No. / Date      | QIR-2025-0233 / 2025-08-19                                                   |
| Part Number            | ACT-SSD-2875-03                                                              |
| Part Description       | SLEEVE, INNER, SHIFTING (QPQ nitrided)                                       |
| Work Order / PO        | WO-25-1799 / Stock                                                           |
| Lot / Heat-Batch       | L25-112 / 8Q4412                                                             |
| Lot Qty / Sample Qty   | 25 / 5 (per ACT-QP-08)                                                       |
| Serial Numbers         | S28-25-0501 through -0505                                                    |
| Material               | AISI 4140, QPQ salt bath nitride, case 0.008-0.015 in.                       |
| Inspected By           | D. Whitfield                                                                 |
| Reviewed / Approved By | S. Nguyen                                                                    |
| Applicable Specs       | API 11D1 / API 19V (as applicable), NACE MR0175/ISO 15156, ASTM E709, API 5B |

## Dimensional & Attribute Results

Actuals are the worst-case (closest-to-limit) reading across the sample unless noted. Instruments are calibrated per ACT-CAL-01; calibration records on file.

| Char. | Feature / Characteristic        | Nominal | Tol +  | Tol -   | Actual (worst) | Instrument           | A/R |
|-------|---------------------------------|---------|--------|---------|----------------|----------------------|-----|
| D1    | Sleeve OD (seal surface)        | 2.870   | +0     | --0.002 | 2.872          | Micrometer M-212     | ACC |
| D2    | Sleeve ID                       | 2.313   | +0.005 | -0      | 2.316          | Bore gauge B-108     | ACC |
| D3    | X-profile groove depth          | 0.140   | +0.005 | --0.005 | 0.145          | Profile gauge XP-2   | ACC |
| D4    | X-profile groove width          | 0.375   | +0.004 | --0.004 | 0.379          | Profile gauge XP-2   | ACC |
| D5    | Detent groove dia               | 2.798   | +0.003 | --0.003 | 2.801          | Groove mic GM-2      | ACC |
| D6    | Surface finish seal OD (Ra uin) | 12.0    | +4.0   | --6.0   | 18.0           | Profilometer P-52    | ACC |
| D7    | Case hardness (HR15N)           | 88.0    | +4.0   | --2.0   | 91.4           | Hardness tester HR-3 | ACC |
| D8    | Stroke length (assembled check) | 2.750   | +0.015 | --0.015 | 2.765          | Height gauge H-14    | ACC |

Result: ACCEPT

## Measuring & Test Equipment Used

| Instrument           | Cal. Record | Cal. Due   | Status                        |
|----------------------|-------------|------------|-------------------------------|
| Micrometer M-212     | CAL-2616    | 2025-12-22 | In calibration at time of use |
| Bore gauge B-108     | CAL-9939    | 2026-07-21 | In calibration at time of use |
| Profile gauge XP-2   | CAL-9082    | 2026-01-08 | In calibration at time of use |
| Groove mic GM-2      | CAL-8478    | 2026-06-09 | In calibration at time of use |
| Profilometer P-52    | CAL-3785    | 2025-05-26 | In calibration at time of use |
| Hardness tester HR-3 | CAL-4897    | 2025-04-23 | In calibration at time of use |
| Height gauge H-14    | CAL-6082    | 2025-03-18 | In calibration at time of use |

## Sampling & Methods

Sampling per ACT-QP-08 (ANSI/ASQ Z1.4 basis). Dimensional measurements at 68-72 degF after minimum 4 hr thermal soak. Surface finish per ASME B46.1. Hardness per ASTM E18 (HRC/HRB) or ASTM E140 conversion where noted. Threads gauged per API 5B with certified working gauges. Magnetic particle inspection, where applied, per ASTM E709 using wet fluorescent method, 100-125 V/in. field.

## Inspector Notes

All five sampled sleeves conform. QPQ case depth verified on sacrificial coupon: 0.011 in. Shift force bench check on S28-25-0501: detent release 355 lbf (spec 350 +/- 50).

## Sign-off

| Role                    | Name         | Signature               | Date       |
|-------------------------|--------------|-------------------------|------------|
| QC Inspector (Level II) | D. Whitfield | /signed electronically/ | 2025-08-19 |
| QA Manager              | S. Nguyen    | /signed electronically/ | 2025-08-19 |